SCH3U UNIT 1: MATTER AND BONDING

NAME: ____________________ DATE: ________________________

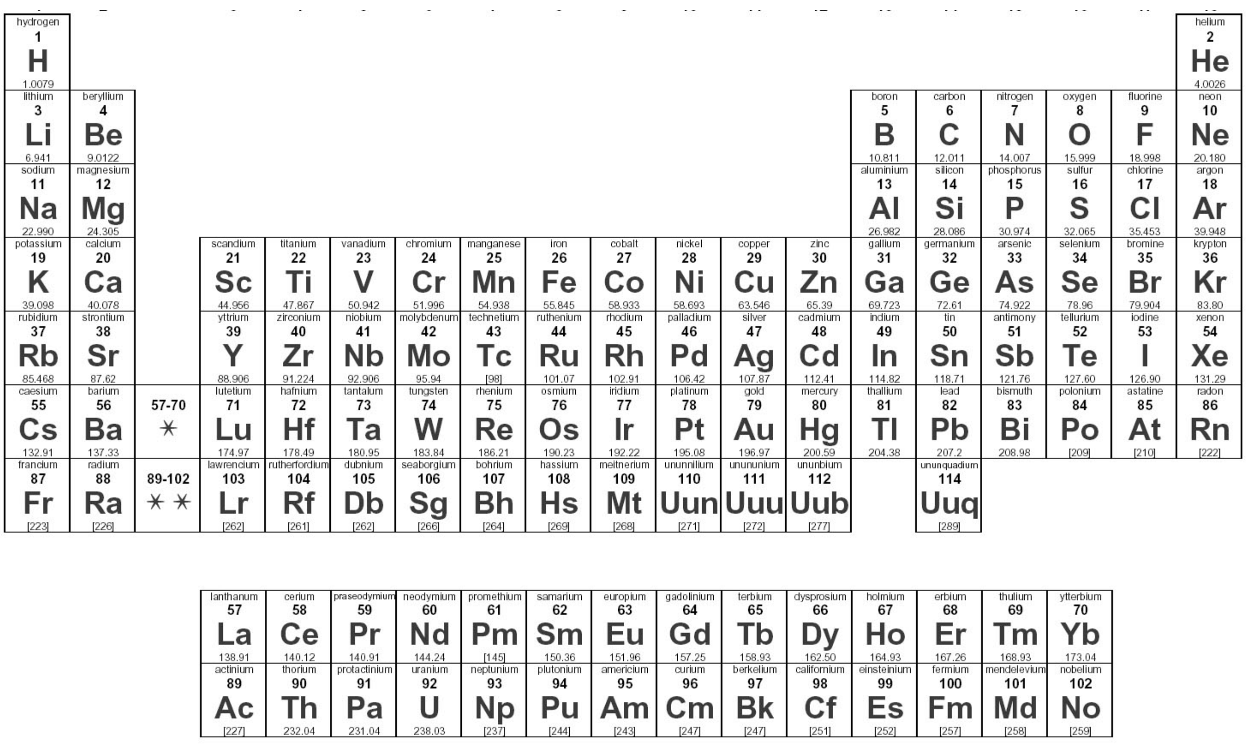
Periodic Trends Assignment

Part A

Using colours and/or shading, Indicate the following:

- Label Hydrogen
- Location of the metals, metalloids, non-metals
- The following families: alkali metals, alkaline earth metals, transition metals, halogens, noble gases, lanthanides (rare earth metals), and actinides
- Label the groups I to VIII

Be sure to include a key, if necessary.



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Part B

Indicate the following trends on the periodic table below. You may use arrows, diagrams, differences in shading and/or symbols. Be sure to include a key, if necessary.

For each property indicate changes 1) across a period AND
2) within a group

- Atomic Radius
- Ionization Energy
- Electron Affinity
- Electronegativity
- Reactivity in a group of non-metals
- Reactivity in a group of metals

hydrogen 1 H 1.0079																		helium 2 He 4.0026
lithium 3 Li 6.941	beryllium 4 Be 9.0122												boron 5 B 10.811	carbon 6 C 12.011	nitrogen 7 N 14.007	oxygen 8 O 15.999	fluorine 9 F 18.998	neon 10 Ne 20.180
sodium 11 Na 22.990	magnesium 12 Mg 24.305												aluminium 13 Al 26.982	silicon 14 Si 28.086	phosphorus 15 P 30.974	sulfur 16 S 32.065	chlorine 17 Cl 35.453	argon 18 Ar 39.948
potassium 19 K 39.098	calcium 20 Ca 40.078		scandium 21 Sc 44.956	titanium 22 Ti 47.867	vanadium 23 V 50.942	chromium 24 Cr 51.996	manganese 25 Mn 54.938	iron 26 Fe 55.845	cobalt 27 Co 58.933	nickel 28 Ni 58.693	copper 29 Cu 63.546	zinc 30 Zn 65.39	gallium 31 Ga 69.723	germanium 32 Ge 72.61	arsenic 33 As 74.922	selenium 34 Se 78.96	bromine 35 Br 79.904	krypton 36 Kr 83.80
rubidium 37 Rb 85.468	strontium 38 Sr 87.62		yttrium 39 Y 88.906	zirconium 40 Zr 91.224	niobium 41 Nb 92.906	molybdenum 42 Mo 95.94	technetium 43 Tc [98]	ruthenium 44 Ru 101.07	rhodium 45 Rh 102.91	palladium 46 Pd 106.42	silver 47 Ag 107.87	cadmium 48 Cd 112.41	indium 49 In 114.82	tin 50 Sn 118.71	antimony 51 Sb 121.76	tellurium 52 Te 127.60	iodine 53 I 126.90	xenon 54 Xe 131.29
caesium 55 Cs 132.91	barium 56 Ba 137.33	57-70 *	lutetium 71 Lu 174.97	hafnium 72 Hf 178.49	tantalum 73 Ta 180.95	tungsten 74 W 183.84	rhenium 75 Re 186.21	osmium 76 Os 190.23	iridium 77 Ir 192.22	platinum 78 Pt 195.08	gold 79 Au 196.97	mercury 80 Hg 200.59	thallium 81 Tl 204.38	lead 82 Pb 207.2	bismuth 83 Bi 208.98	polonium 84 Po [209]	astatine 85 At [210]	radon 86 Rn [222]
francium 87 Fr [223]	radium 88 Ra [226]	89-102 **	lawrencium 103 Lr [262]	rutherfordium 104 Rf [261]	dubnium 105 Db [262]	seaborgium 106 Sg [266]	bohrium 107 Bh [264]	hassium 108 Hs [269]	meitnerium 109 Mt [268]	ununnilium 110 Uun [271]	unununium 111 Uuu [272]	ununbium 112 Uub [277]		ununquadium 114 Uuq [289]				

lanthanum 57 La 138.91	cerium 58 Ce 140.12	praseodymium 59 Pr 140.91	neodymium 60 Nd 144.24	promethium 61 Pm [145]	samarium 62 Sm 150.36	europium 63 Eu 151.96	gadolinium 64 Gd 157.25	terbium 65 Tb 158.93	dysprosium 66 Dy 162.50	holmium 67 Ho 164.93	erbium 68 Er 167.26	thulium 69 Tm 168.93	ytterbium 70 Yb 173.04
actinium 89 Ac [227]	thorium 90 Th 232.04	protactinium 91 Pa 231.04	uranium 92 U 238.03	neptunium 93 Np [237]	plutonium 94 Pu [244]	americium 95 Am [243]	curium 96 Cm [247]	berkelium 97 Bk [247]	californium 98 Cf [251]	einsteinium 99 Es [252]	fermium 100 Fm [257]	mendelevium 101 Md [258]	nobelium 102 No [259]

Part C

Using your knowledge of atomic radius, electronegativity, and ionization energy explain **fully** the trends in reactivity in metals and non-metals. This should be done on a separate piece of paper and be approximately ½ page to a page in length.

NAME: ____________________ DATE: ________________________

Part D

Graph the following sets of data to show the trends that exist in atomic radii and ionization energies. You should have two separate graphs: A) Atomic Number vs Atomic Radius
B) Atomic Number vs Ionization Energy

Atomic Number (Z#)	Element	Atomic Radius	First Ionization Energy (kJ/mol)
1	H	0.032	1312
2	He	0.031	2372
3	Li	0.123	520
4	Be	0.090	899
5	Be	0.082	801
6	C	0.077	1086
7	N	0.075	1402
8	O	0.073	1314
9	F	0.072	1681
10	Ne	0.071	2081
11	Na	0.154	496
12	Mg	0.136	738
13	Al	0.118	578
14	Si	0.111	786
15	P	0.106	1012
16	S	0.102	1000
17	Cl	0.099	1251
18	Ar	0.098	1521
19	K	0.203	419
20	Ca	0.174	590
21	Sc	0.144	631
22	Ti	0.132	658
23	V	0.122	650
24	Cr	0.118	653
25	Mn		
26	Fe	0.117	759
27	Co	0.116	758
28	Ni	0.115	737
29	Cu	0.117	745
30	Zn	0.125	906
31	Ga	0.126	579
32	Ge	0.122	762
33	As	0.120	947
34	Se		
35	Br	0.114	1140
36	Kr	0.112	1351